

09 Thyroid Synthetic-Lethal Vulnerability Map (Plan)

Forward-looking translational paper that takes the DM1 / RAI-lineage-collapsed state defined in Paper 1 as a state-of-vulnerability and asks: "given this state, what genes/drugs kill the tumor selectively, and which compensatory partners must we co-target?" Ruppin-style in spirit, thyroid-tuned and DM1-anchored.

Status: STRATEGIC DESIGN ONLY

No data download · no analysis · no GPU · no manuscript

G0 conditional on Paper 1 in print

Author: **Seungho Cook**

Earliest start: 2026-06-16

Date: 2026-05-04

I. 한 줄 결론

GO, conditional on Paper 1 in print. Paper 9 = forward-looking translational paper. DM1 state 를 state-of-vulnerability 로 잡고 "이 state 에서 무엇이 selective 하게 죽이는가 + 어떤 보상 partner 를 같이 눌러야 하는가" 를 묻는다. Ruppin-style 인 *conceptually* 하지만 thyroid-tuned · state-anchored · transcription-based – ISLE/SELECT 문장 / signature / pair table 은 lift 안 함. 본 페이지는 design only – data 다운로드 없음, 분석 없음, GPU 없음, 본문 없음.

STATUS

DESIGN

No data, no analysis,
no manuscript

EARLIEST START

2026-06-16

Post-marathon, after
Paper 1 in print

TARGET
CLASSES

8

+ 1
carved-
out
non-SL
(TROP2
ADC)

EXECUTION
SPRINT

12
weeks

Step A → Step
F · gated
checkpoints

WHY A REAL PAPER

Paper 1
defines state
but does not
prosecute
drug
vulnerability.
Pan-cancer SL
engines under-
sample
thyroid; DM1
lineage
collapse +
compensatory
rewiring is
precisely
where SL

RISKS

thyroid-
specific
CRISPR depth
in DepMap is
shallow;
PRISM/
GDSC drug
coverage
sparse; SL
inference
statistically
delicate.
Manageable,
requires
honest
framing.

DEFENSE

state-
anchored, not
mutation-
anchored.
Borrow
conceptual
scaffolding
only. Cite
Ruppin
generously;
never claim
equivalence.

DECISION

Wait for Paper
1 in print +
marathon
close + Yu
sign-off. Then
12-week
sprint with
gated GO/
NO-GO at
each step.

theory has the
most leverage.

2. 비전공자 배경

합성치사 (synthetic lethality, SL) = 두 유전자 중 하나만 망가지면 세포가 사는데 둘 다 망가지면 세포가 죽는 관계. 암이 이미 한쪽 망가져 있으면, 나머지 한쪽을 약으로 누르면 정상세포는 살고 암세포만 선택적으로 죽는다. SL 기반 항암제의 logic.

본 paper 의 출발점은 Paper 1 의 **DM1 (dark matter 1)** state. 갑상선암 안에서 갑상선 lineage program (FOXE1/NKX2-1/PAX8/HHEX) 이 collapse 한 상태. 이 state 가 STAT3/AP-1/DNMT 같은 보상 회로 활성화를 동반한다 – *lineage* 가 무너지면서 동시에 살아남기 위해 다른 회로를 켜다. SL 이론이 가장 잘 먹히는 regime: lost redundancy → narrow survival corridor.

Paper 9 는 이 corridor 안의 어느 노드가 약을 누를 만한 약한 고리인지 묻는다. 답은 **state-conditional** – "갑상선암 일반" 이 아니라 "DM1-like state 의 갑상선암" 에서만 켜지는 dependency. 본 페이지는 그 검정 plan 의 설계도 다 – 실제 검정은 post-marathon.

2.I 용어 정리

TERM	풀어쓴 설명	이 PLAN 에서의 역할
DM1 / RAI-lineage	Paper 1 정의. lineage-collapsed thyroid transcriptional state.	state-of-vulnerability anchor.
SL / Synthetic Lethality	한쪽 손상 + 다른 쪽 표적 = 선택적 사멸.	main therapeutic logic.
SR / Synthetic Rescue	partner 의 up-regulation 이 표적 손실의 fitness 손해를 보상해서 저항성 만든다.	resistance prediction.
State-conditional dependency	특정 transcriptional state 에서만 커지는 CRISPR essentiality.	thyroid-specific 적응의 핵심.
Ruppin-style	Eytan Ruppin lab 의 ISLE / SELECT 등 transcriptome-guided SL inference 계열.	conceptual scaffolding (text/data lift 금지).
TROP2 / TACSTD2	tumor-cell-surface marker. ADC payload delivery 표적.	명시적 carve-out: SL 아님, parallel translational opportunity.

3. 전체 논리 흐름

PAPER 1 (ANCHOR)

DM1 state defined

8-gene RAI compact
readout + TF-collapse score.
lineage-collapsed thyroid
의 reproducible signature.

→ asks

PAPER 9 (PLAN)

"Given DM1, what kills it?"

state-anchored SL
inference + drug map +
cohort validation. Post-
marathon 12-week sprint.

Q1 - 왜 PAPER 인가, 메모인가?

Paper 1 은 state 정의만 함. drug
vulnerability prosecute 안 함. SL/
SR map + 임상 cohort validation
= 별도 paper.

Q2 - 왜 PAPER 1 다음인가, PARALLEL 아닌가?

citable state 가 아니면 reviewer 가
state 정의 자체를 시비. Paper 1 in
print 가 anchor citation 으로 작
동.

Q3 - 왜 RUPPIN-STYLE 인가?

transcriptome-guided SL
inference 가 mutation-anchored
보다 mutation-poor / RAI-
refractory thyroid 에 적합.

Q4 - 왜 ISLE/SELECT 직접 쓰지 않 는가?

thyroid 가 pan-cancer SL pair 들
에서 under-represented. state-
anchored 방식이 thyroid-specific
signal 을 살린다.

Q5 - 가장 큰 RISK?

thyroid CRISPR depth 부족.
cross-tissue borrowing + thyroid
으로 always validate-back 으로 완
화.

FINAL - WAIT + 12-WEEK SPRINT.

Paper 1 in print → marathon
close → Yu sign-off → Step A. 매
step 에 GO/NO-GO.

§ 1 — Executive verdict

GO, conditional on Paper 1 in print.

Paper 9 is a **forward-looking translational paper** that takes the DM1 / RAI-lineage-collapsed state defined in Paper 1 as a **state-of-vulnerability** and asks: "given this state, what genes/drugs kill the tumor selectively, and which compensatory partners must we co-target?" The design is Ruppin-style in spirit (transcriptome-guided synthetic lethality / synthetic rescue, population-level co-occurrence patterns, cross-cohort transferability) but specifically thyroid-tuned and DM1-anchored.

Why this is a real paper and not an extension memo:

- Paper 1 defines a state but does not prosecute drug vulnerability.
- Existing pan-cancer SL inference engines (ISLE, SELECT, etc.) systematically under-sample thyroid; their inferred pairs are not directly transferable.
- DM1-like dedifferentiation (FOXE1/NKX2-1/PAX8/HHEX collapse + STAT3/AP-1/DNMT activation) is an unusual transcriptional state – lineage collapse plus compensatory rewiring – and is precisely the regime where SL theory has the most leverage (lost redundancy → narrow survival corridor).

Risks: lack of thyroid-specific CRISPR depth in DepMap; PRISM/GDSC drug coverage is sparse for thyroid; SL inference is statistically delicate. These are *manageable* but require honest framing (§ 12).

§ 2 — Why Paper 9 exists after Paper 1, not in parallel

PAPER	ASKS	ESTABLISHES
Paper 1	What is DM1?	A reproducible transcriptional state + 8-gene readout
Paper 2	How does Hashimoto-overlap PTC project onto DM1 morphologically/ biologically?	DM1 visibility on H&E + immune layer
Paper 3	Why is the dark thyroid cancer ICI-vulnerable?	Immunological vulnerability axis
Paper 4 (backlog)	Korean GD HLA Pan-Asian	independent immunogenetic line
Paper 9	Given DM1, what kills it?	Druggable / wet-lab-testable vulnerability map

Sequencing rationale:

- Paper 9 needs the DM1 state to be *citable*, not "in prep". Otherwise reviewers fight the state definition instead of the SL claim.
- Paper 1 in print = anchor citation. Paper 9 inherits authority.
- Running Paper 9 in parallel would dilute Paper 1's identity ("8-gene readout becomes the basis for a drug map") – better as the *next* paper's headline.

§ 3 — Ruppín-style conceptual template (used as scaffolding, not as text)

We borrow concepts and method-class, not language. The conceptual scaffolding is publicly described in the field and may be re-implemented as long as we do not lift sentences, claim independent priority over their algorithms, or imply equivalence to ISLE/SELECT outputs.

Concepts we adopt:

1. **Synthetic lethality (SL) as a population-level statistical signal.** Pairs (A, B) where co-loss is depleted in tumors that survive (because co-loss kills the cell) – operationalized via co-occurrence / co-expression / survival statistics.
2. **Synthetic rescue (SR).** Partner whose up-regulation rescues fitness when the target is lost; predicts resistance.
3. **Transcriptome as therapy-selection layer.** Move beyond actionable mutations: use gene-expression state to predict drug response, especially in mutation-poor tumors (which DM1 partially is).
4. **Cross-data triangulation.** SL/SR candidates must show a coherent signal across (i) cell-line dependency, (ii) drug response, (iii) patient outcome – not just one.

Concepts we explicitly do not adopt or claim:

- We do not claim ISLE-equivalent or SELECT-equivalent inference.
- We do not lift their precomputed SL pair lists as ground truth.
- We do not benchmark our pipeline against theirs as a horse-race; we treat them as inspiration, not adversary.

If Ruppín-tools are run *for sanity check* (recommended but optional), they are cited as third-party reference, not as our pipeline.

§ 4 — Thyroid-specific adaptation

Why thyroid breaks pan-cancer SL out of the box:

- Thyroid lines in DepMap are few (≈ 12 named, fewer with full CRISPR depth); pan-cancer SL pairs are dominated by signal from over-represented lineages (breast, colon, lung).
- Thyroid mutation density is low; co-mutation-based SL inference is anemic.
- Thyroid lineage TFs (PAX8, FOXE1, NKX2-1) are not standard SL nodes in pan-cancer maps.

Adaptations:

- 1 State-conditioned SL.** Instead of asking "is gene X SL with Y pan-cancer", ask "is X SL with Y *given a DM1-like transcriptional context*". Define DM1-like state at the cell-line level using the 8-gene compact readout + TF-collapse score, then condition SL inference on that state.
- 2 Lineage-imputed pseudobulk.** Augment thin thyroid cell-line panel with thyroid PDX/organoid expression where available, treated as held-out validation, not training.
- 3 Cross-tissue borrowing for power.** Where the DM1-like transcriptional state appears in non-thyroid lineages (e.g., dedifferentiated neuroendocrine, lineage-plastic prostate), borrow strength for SL inference but always validate the hit *back* in thyroid context.
- 4 TF-collapse as the SL anchor, not driver mutation.** This is the single biggest conceptual departure from canonical Ruppert work, which is mutation-anchored. We are state-anchored.

§ 5 — Main hypothesis (single sentence + decomposition)

H0: DM1-like thyroid tumors carry a non-redundant set of synthetic-lethal vulnerabilities arising from lineage collapse (FOXE1/NKX2-1/PAX8/HHEX loss) and compensatory rewiring (STAT3/AP-1/DNMT activation), and these vulnerabilities are predictable from the 8-gene compact readout without requiring driver-mutation profiling.

Decomposition:

- **H1 (state):** DM1-like state is reproducibly callable in CCLE.
- **H2 (dependency):** A subset of CRISPR dependencies in DepMap is significantly enriched in DM1-like lines.
- **H3 (compensation):** A subset of these dependencies has a synthetic-rescue partner whose up-regulation predicts resistance.
- **H4 (drug):** Drug response in PRISM/GDSC/CTRP recapitulates H2 for druggable targets.
- **H5 (outcome):** In TCGA-THCA and GSE76039, high target-pathway expression in DM1-classified tumors stratifies survival – motivating but not proving therapeutic benefit.

§ 6 — Candidate target classes

For each: rationale (DM1-link), prior thyroid evidence, drug class, what would falsify, SL-vs-non-SL classification.

6.1 NAMPT / NAD salvage

- **Rationale:** STAT3-high + high transcriptional output → high NAD turnover. NAMPT is the rate-limiter of NAD salvage. Cancers with metabolic plasticity often become NAD-addicted.
- **Drugs:** FK866 (APO866), KPT-9274, OT-82.
- **SL?** Yes, candidate SL with PARP-pathway / OXPHOS partners; SR with NAPRT (NAD de novo arm) – NAPRT-low + NAMPT-high is the classic vulnerability window.
- **Falsifier:** if NAPRT is uniformly high in DM1 lines, vulnerability vanishes.

6.2 STAT₃ / JAK

- **Rationale:** Direct readout of the DM1 inflammatory-rewiring signature.
- **Drugs:** ruxolitinib (JAK1/2), TYK2-selective (deucravacitinib-class), STAT3-selective (TTI-101, napabucasin caveat).
- **SL?** Often *not* SL – STAT3 inhibition alone gives partial response. Real SL targets are likely **upstream cytokine receptors (IL6R, OSMR)** or **parallel TFs (NFκB)** in DM1 context.
- **Falsifier:** STAT3 single-agent inhibitor sensitivity does not separate DM1-like vs. non-DM1 lines.

6.3 DNMT / epigenetic repression

- **Rationale:** DNMT activation in DM1 → hypermethylation of lineage TFs (FOXE1 promoter is a known target). DNMT inhibitors may *reactivate* lineage program.
- **Drugs:** decitabine, guadecitabine; combination with HDACi (vorinostat) or LSD1i.

- **SL?** Conceptually different: this is **lineage-rewiring**, not pure SL. We classify it as a **redifferentiation vulnerability** and frame separately.
- **Falsifier:** DNMTi does not restore FOXE1/PAX8/NKX2-1/HHEX expression in DM1-like cell lines.

6.4 LYN / SRC-family kinases

- **Rationale:** Compensatory tyrosine-kinase signaling in mutation-poor dedifferentiated thyroid; SFK activation is reported in advanced/anaplastic thyroid cancer.
- **Drugs:** dasatinib (broad SFK), bosutinib, saracatinib.
- **SL?** Plausible SL partner with PI3K/AKT or with STAT3.
- **Falsifier:** SFK dependency in DepMap is not enriched in DM1-like lines.

6.5 KCNN4 (KCa3.1)

- **Rationale:** Calcium-activated K⁺ channel; up-regulated in some dedifferentiated solid tumors; reported in immune-tumor crosstalk.
- **Drugs:** senicapoc, TRAM-34 (tool compound).
- **SL?** More likely a *modifier* than a hard SL; included because of CRISPR-screen reproducibility in adjacent contexts.
- **Falsifier:** No DepMap dependency signal at all in DM1-like context.

6.6 DNA damage / PARP

- **Rationale:** Lineage plasticity often correlates with replication stress and BRCA-ness signatures.
- **Drugs:** olaparib, talazoparib, niraparib; ATR inhibitors (ceralasertib).
- **SL?** Yes if replication-stress signature is elevated in DM1; SR partner is BRCA1/2 / HR pathway intact tumors that resist.
- **Falsifier:** No HRD signature / no replication-stress markers (CHEK1, ATR, RPA pathway) elevated in DM1-like tumors.

6.7 Redox / OXPHOS-glycolysis compensation

- **Rationale:** Lineage collapse often forces metabolic reprogramming. DM1 may rely on either OXPHOS (mitochondrial) or compensatory glutamine/glycolysis.
- **Drugs:** IACS-010759 (OXPHOS), CB-839 (glutaminase), 2-DG (research tool).
- **SL?** Complementary metabolic SL pair: GLS1 (glutaminase) ↔ MYC, or OXPHOS ↔ glycolytic capacity (LDHA/HK2).
- **Falsifier:** Metabolic flux signature in DM1 is no different from generic high-proliferation thyroid lines.

6.8 TROP2 — explicit non-SL ADC vulnerability (carved out)

Important framing. TROP2 is a tumor-cell-surface marker, not a synthetic-lethal partner. Mixing the two categories has been a recurring failure mode in similar papers and we will not.

- **Drugs:** sacituzumab govitecan (ADC, payload SN-38).
- **SL? NO.** This is *expression-level vulnerability* for ADC delivery. We include it in the paper as a **parallel translational opportunity**, explicitly distinguished from the SL-map.
- **What we say:** "TROP2-high DM1 tumors are candidates for ADC therapy, by an orthogonal mechanism to the synthetic-lethal targets above."

§ 7 — Data needed later (no download now)

Acquisition is **post-marathon**. List for planning only.

SOURCE	USE	ACQUISITION
DepMap CRISPR (DependencyMap, Broad)	Per-gene dependency in cell lines, conditioned on DM1-like state	Public; figshare / depmap.org
CCLE expression	Define DM1-like cell lines	Public
PRISM Repurposing	Drug-response across ~5,000 compounds, broad cell-line panel	Public
GDSC1/GDSC2	Drug-response, ~700 compounds, deeper IC50 fits	Public
CTRP v2	Redundant drug-response cross-check	Public
TCGA-THCA RNA + mutation + clinical	Patient-cohort validation; outcome stratification	dbGaP/UCSC Xena (already locally available)
GSE76039	Advanced-disease replication; we already use this in Paper 1	Already on disk
Thyroid PDX expression panels (where published)	Held-out validation, not training	Selective public sets
Organoid datasets (where available)	Wet-lab grounding for top hits	Selective public

What we will NOT pull until Paper 1 is in print: DepMap, PRISM, GDSC, CTRP, organoid sets – all deferred. This document reserves the slot, it does not start the work.

§ 8 — Analysis plan (when work begins)

Sequenced, not started. Each step has an explicit GO/NO-GO criterion.

- 1 Step A – Define DM1-like state at cell-line level.** Score CCLE lines on the 8-gene compact readout + TF-collapse composite (FOXE1/NKX2-1/PAX8/HHEX). Calibrate against thyroid lines we already trust as DM1-like (anaplastic, poorly differentiated). *GO if ≥ 6 thyroid lines + ≥ 30 cross-tissue lines hit DM1-like threshold.* NO-GO recovery: relax cross-tissue threshold; re-anchor on TF-collapse only.
- 2 Step B – Compute target dependency.** For each gene, contrast CRISPR dependency score in DM1-like vs. non-DM1-like lines. Multiple-testing correction across the genome. Stratify by lineage to detect thyroid-specific vs. shared. *GO if any of {NAMPT, NAPRT, STAT3, JAK1/2, DNMT1, DNMT3B, LYN, FYN, SRC, GLS1, ATR} shows DM1-conditioned enrichment.*
- 3 Step C – Infer SL / SR pairs.** For top dependencies, identify partners whose co-loss / co-up-regulation modulates dependency. Cross-validate against Ruppin pan-cancer SL pair lists *as a sanity probe only* – not as ground truth. *GO if at least 2 partner-gated SL pairs survive.*
- 4 Step D – Rank druggable vulnerabilities.** Cross-reference Step B/C hits with PRISM/GDSC drug-target maps. Score by: (i) effect size in DM1-like lines, (ii) target druggability, (iii) clinical-stage drug availability, (iv) thyroid-cell-line confirmation. Output: ranked top-10 vulnerability list.
- 5 Step E – Validate in patient cohorts.** In TCGA-THCA: stratify by DM1 classifier; test whether high target-pathway score associates with survival, recurrence, RAI-refractoriness flags. Replicate in GSE76039 (advanced-disease, smaller N). *GO if at least 1 target shows concordant direction across both.*



Step F – Stage results into figure plan F1–F6.

§ 9 — Figure plan F1–F6

- **F1 – Concept.** Schematic: DM1 state → lost redundancy → narrowed compensatory pathways → druggable nodes. Adapted from Ruppin-style SL/SR concept; visually distinct from any published ISLE/SELECT figure.
- **F2 – DM1-like state in CCLE.** Heatmap of 8-gene + TF-collapse score across lines; thyroid lines flagged; cross-tissue DM1-like lines marked.
- **F3 – Genome-wide CRISPR dependency, DM1 vs. non-DM1.** Volcano. Highlight top 20 hits; color-code by target class (NAD, JAK/STAT, epigenetic, SFK, metabolic, DNA damage).
- **F4 – Top SL/SR pairs.** Ten-row mini-grid: target gene × partner gene, with effect sizes for SL and SR axes.
- **F5 – Drug response (PRISM / GDSC).** IC50 distributions for top druggable targets, DM1-like vs. non-DM1-like. Add NAMPT/NAPRT, JAK, DNMT, SFK panels.
- **F6 – Patient validation (TCGA-THCA + GSE76039).** Survival stratification by DM1 + target-pathway score. Forest plot for hazard ratios.

Supplementary: per-target sensitivity panels; cell-line list with DM1 score; SL/SR statistical tables; sanity-check overlap with Ruppin pan-cancer SL lists (as reference, not benchmark).

§ 10 — Wet-lab validation options (not committed; describes the landscape)

TIER 1 – MINIMUM TO DEFEND THE PAPER

- siRNA / CRISPR knock-down in 2–3 thyroid cell lines (one DM1-like, one non-DM1-like control) for the top 2 SL targets.
- Drug-response curves (IC₅₀ + viability) for matching small-molecule inhibitors.

TIER 2 – STRONG BUT OPTIONAL

- Thyroid organoid panel (if accessible via collaborator) – drug response on top 3 hits.
- Combination experiments (target + SR partner inhibitor) to demonstrate the SR rescue logic.

TIER 3 – BEYOND FIRST PAPER, FOR FOLLOW-UP

- PDX in vivo for top 1 hit.
- Patient-derived organoid bank cross-screen.

DECISION

Paper is *defendable* with Tier 1; Tier 2 is a stretch for first round; Tier 3 belongs to a follow-up. Cost / collaborator question: thyroid wet-lab access. Open. If no wet-lab partner, paper still publishable as **computational + patient-cohort** vulnerability map, framed as hypothesis-generating, in a translational journal (JCI Insight, Mol Cancer Therap, NPJ Precision Oncology) rather than a flagship outlet.

§ II — Business / IP angle

- **Composition-of-matter:** not patentable for repurposed approved drugs (NAMPT inhibitors in trials, ruxolitinib, decitabine, dasatinib, olaparib, IACS-010759, CB-839 – all third-party).
- **Method-of-use / method-of-treatment:** potentially patentable: "method of selecting thyroid cancer patients for [target-class] therapy by measuring the 8-gene DM1 readout + TF-collapse score." This is the actionable IP slot.
- **Companion diagnostic:** the 8-gene readout itself is the most defensible asset; CDx claims tied to specific drug classes are filable separately.
- **Trial design:** umbrella / basket trial design where the DM1 score routes patients to one of 3–4 target arms (NAD, JAK/STAT, epigenetic, ADC). Pitchable to industry partners after Paper 9 publishes.
- **Defensive citation:** cite Ruppin-group methods as the conceptual prior art *generously* to avoid any infringement appearance and to position our work as thyroid-specific, state-anchored, clinically translatable.

§ 12 — What not to claim

Hard rules. Violating any of these wrecks reviewer trust.

1. Do **not** claim direct equivalence with ISLE/SELECT outputs. We are conceptually inspired, not replicating.
2. Do **not** claim clinical efficacy. Patient-cohort survival stratification is hypothesis-generating, not proof of therapeutic benefit.
3. Do **not** claim DM1-readout replaces driver-mutation profiling. Frame as **complementary**, especially in mutation-poor / RAI-refractory settings.
4. Do **not** mix TROP2-ADC vulnerability with synthetic lethality. They are different mechanistic categories. Carved out in § 6.8.
5. Do **not** over-extend to Hashimoto-overlap or Graves' subtypes. That is Paper 2 / Paper 4 territory; Paper 9 is specifically the **DM1 dark-matter axis**.
6. Do **not** use the H&E-DM1 angle. NO-GO per existing standing decision. Stay in molecular / drug-response / outcome space.
7. Do **not** claim wet-lab validation we have not done. If Tier 1 is missing, label the paper "computational + cohort-validated, wet-lab pending" explicitly in the abstract.
8. Do **not** lift sentences, signature lists, or precomputed pair tables from Ruppin-group papers. Concept yes, text/data no.
9. Do **not** start any of this before Paper 1 is in print and the marathon is closed (2026-06-13 floor).

§ 13 — Why this is Paper 9, not Paper

1 / 2 / 3

QUESTION	ANSWER
Could this be folded into Paper 1?	No. Paper 1 establishes biology; folding in SL inflates scope and invites reviewer mutiny on the state definition.
Could this be Paper 2?	No. Paper 2 is the Hashimoto-overlap PTC paper (HT-isolated per 2026-05-04 Yu decision). Different cohort, different question.
Could this be Paper 3?	No. Paper 3 is the ICI-vulnerability dark thyroid cancer paper (FROZEN bundle). Synthetic lethality is a different therapeutic axis; mixing them dilutes both.
Why number 9 specifically?	Reserves a slot in the long-range pipeline. Papers 5–8 are reserved for [paper-numbering doc owner-defined slots]; Paper 9 is the first translational follow-up to the DM1 axis. The number is conventional, not literal – reassign if numbering changes.
Why not just publish target list as a follow-up letter?	Letter-format does not support the figure-and-cohort weight of F1–F6 + supplements. Full article is the right vehicle.

§ 14 — Twelve-week future execution plan (post-marathon)

Floor: cannot start before manuscript marathon closes (2026-06-13).

Earliest practical start: 2026-06-16. Twelve-week sprint plan, each week is a checkpoint, not a deadline.

WK	PHASE	OUTPUT
1	DepMap + CCLE pull, integrity checks	Local mirror, version-pinned
2	Define DM1-like state in CCLE; score thyroid lines	DM1 score table, cell-line ranking
3	Genome-wide DM1-conditioned dependency analysis	Volcano (F3 draft)
4	SL / SR pair inference	Pair table, sanity cross-check vs. Ruppin lists
5	PRISM / GDSC pull and integration	Drug-response matrix
6	Drug-response stratification by DM1; F5 draft	Per-target IC50 panels
7	TCGA-THCA + GSE76039 patient validation	Survival forest plots (F6 draft)
8	Cross-checks: lineage, batch, confounders	Sensitivity tables
9	Figure assembly F1–F6, supplements	Figure pack v1
10	Draft writing (intro, results, discussion)	Manuscript v0
11	Internal review with Yu / collaborator(s); revise	Manuscript v1
12	Submission package: cover letter, COI, key resources	Submitted

Wet-lab Tier 1 is a parallel track running weeks 4–12 if a collaborator is available; otherwise the paper is submitted as computational + cohort-validated and Tier 1 is the response-to-reviewer plan.

Stop conditions.

- If Step A (DM1-like state in CCLE) fails: pivot to bulk pseudobulk validation only and reframe scope. Do not fabricate state membership.
- If Step C (SL pairs) yields zero hits: paper still publishable as a *negative* result + ranked dependency list, but in a lower-tier venue. Decide explicitly at week 4 checkpoint.
- If wet-lab Tier 1 fails for top hits: replace top hits with rank-2/3 vulnerabilities, do not bury the failure.

15. Claim boundary

이 plan 자체는 분석을 시작하지 않은 design document. claim 은 *plan* 의 spec 에 한정된다.

CLAIM	STATUS
Strategic plan + 12-week sprint design	YES
Conceptual scaffolding (Ruppin-style SL/SR adapted to thyroid)	YES
Candidate target classes + drug landscape (8 + 1 carved-out)	YES
Hypothesis decomposition (H1~H5)	DESIGN ONLY
Any analysis result	NO (deferred)
Any data download	NO (deferred)
Any manuscript text	NO (post-Paper 1)
ISLE / SELECT equivalence	NO (concept-only borrow)
Validated synthetic lethality	NO (post-DepMap + post-wet-lab)
Clinical efficacy	NO

Next action gates (in order): 1. Paper 1 in print. 2. Marathon closed ($\geq 2026-06-13$). 3. Yu sign-off on Paper 9 slot. 4. Then and only then: Step A.

I6. 데이터 / 다운로드 . 자매 페이지

This document is design only. No data has been pulled, no analysis run, no GPU launched, no manuscript text written. Paper 1, 2, 3, 4 not touched. Voice-protected sections not authored. Paper 9 / 10 / 11 / 12 chain links below.

← PAPER 9 LANDING

SL companion landing page

PAPER 9 FIRST-PASS →

Local-data-only triangulation results

PAPER 10 ATLAS

17 figures + 9 sortable tables companion

PAPER 11 PANCANCER

DM1 transfer atlas across 33 lineages

PAPER 12 NETWORK

Candidate × candidate co-expression

← PAPER 1 ANCHOR

DM1 / RAI-lineage axis 정의

PHASE B PLATFORM

TROP2 ADC carve-out (non-SL)

PAPER 9 PORTFOLIO CARD

One-page detail card

← HUB INDEX

8-papers portfolio

Status. Strategic design only. No data, no analysis, no manuscript. Anchor doc: `2026_05_04_paper9_synthetic_lethality_ruppin_style_plan.md`. Forbidden in this doc: copying ISLE/SELECT prose, claims, figures, or signatures.

Live URL. `http://40.82.129.113:8012/papers_hub_2026_05_04/paper9_plan.html`